MLOps Research Study

Automated Research Summary Report

This compact report collects the principal configuration, validation, generalization, learning-curve, and permutation-sensitivity outputs produced by the current MLOps run. Detailed CSV/JSON files remain the authoritative source for full results.

Setting	Value
Generated	2026-08-23T11:37:27+03:30
Dataset	G:\Researching\GT_GL_MLops\GLFS_M...
Dataset SHA256	daf22e9b3887a8070242eec567c90206a...
Targets	AC
Grouping enabled	False
Group column	Location_No
Train / test	0.8/0.2
CV splits / repeats	5/3
Classification enabled	True
Class boundaries	{"AC": {"lower": 0.75, "upper": 1...

Analysis availability

Analysis	Status	Files
Data quality	Available	5
Multicollinearity	Available	4
Regression nested CV	Available	48
Regression learning curves	Available	48
Regression permutation	Available	48
Classification nested CV	Available	36
Classification learning curves	Available	36
Classification permutation	Available	36

Selected / representative model results

Task	Target	Feature set	Model	Metric	CV	CV SD	Test	Gap
regression	AC	fs1	XGB	R2	0.500	0.208	0.560	0.409
regression	AC	fs2	DT	R2	0.614	0.325	0.669	0.261
regression	AC	fs3	DT	R2	0.610	0.183	0.500	0.199
regression	AC	fs4	XGB	R2	0.489	0.133	0.556	0.434
classification	AC	fs1	RF	Macro-F1	0.660	0.167	0.831	0.321
classification	AC	fs2	DT	Macro-F1	0.719	0.061	0.843	0.147
classification	AC	fs3	DT	Macro-F1	0.710	0.104	0.892	0.097
classification	AC	fs4	XGB	Macro-F1	0.621	0.168	0.767	0.306

Nested cross-validation

Nested CV is summarized from the saved outer-fold files. Hyperparameter optimization remains inside the inner CV loop in the training programs; the table below reports the outer-fold generalization estimates when available.

Task	Target	Feature set	Model	Outer folds	Metric	Mean	SD	Secondary	Value
classification	AC	fs2	RF	5	Macro-F1	0.712	0.187	Balanced accuracy	0.747
classification	AC	fs1	XGB	5	Macro-F1	0.693	0.167	Balanced accuracy	0.711
classification	AC	fs4	RF	5	Macro-F1	0.686	0.118	Balanced accuracy	0.723
classification	AC	fs3	RF	5	Macro-F1	0.659	0.217	Balanced accuracy	0.691
regression	AC	fs2	GBR	5	R2	0.565	0.249	RMSE	0.246
regression	AC	fs3	RF	5	R2	0.519	0.280	RMSE	0.265
regression	AC	fs1	XGB	5	R2	0.469	0.248	RMSE	0.288
regression	AC	fs4	RF	5	R2	0.446	0.363	RMSE	0.292

Learning-curve analysis

Task	Target	Feature set	Model	Final CV R2	Final CV F1	Final gap	Trend	More-data signal
regression	AC	fs1	XGB	0.500	-	0.420	improving	True
regression	AC	fs1	GBR	0.457	-	0.325	improving	True
regression	AC	fs1	DT	0.441	-	0.363	improving	True
regression	AC	fs1	RF	0.433	-	0.469	improving	True
regression	AC	fs1	ET	0.284	-	0.432	improving	False
regression	AC	fs1	SVR	0.280	-	0.223	improving	False
regression	AC	fs1	ANN	0.259	-	0.388	improving	True
regression	AC	fs1	KNN	0.116	-	0.884	improving	True
regression	AC	fs1	EN	0.104	-	0.279	near plateau	False
regression	AC	fs1	LASSO	0.087	-	0.239	near plateau	False
regression	AC	fs1	RIDGE	0.051	-	0.326	near plateau	False
regression	AC	fs1	MLR	0.015	-	0.415	declining	False
regression	AC	fs2	DT	0.614	-	0.248	improving	True

Task	Target	Feature set	Model	Final CV R2	Final CV F1	Final gap	Trend	More-data signal
regression	AC	fs2	XGB	0.608	-	0.377	improving	True
regression	AC	fs2	GBR	0.581	-	0.341	improving	True
regression	AC	fs2	RF	0.554	-	0.335	improving	True
regression	AC	fs2	ET	0.274	-	0.437	improving	False
regression	AC	fs2	ANN	0.272	-	0.131	improving	True
regression	AC	fs2	SVR	0.270	-	0.274	improving	True
regression	AC	fs2	EN	0.172	-	0.277	near plateau	False
regression	AC	fs2	LASSO	0.157	-	0.320	declining	False
regression	AC	fs2	RIDGE	0.131	-	0.296	improving	False
regression	AC	fs2	MLR	0.112	-	0.368	declining	False
regression	AC	fs2	KNN	0.039	-	0.961	improving	True

Permutation sensitivity / feature importance

Permutation sensitivity is computed by perturbing one feature at a time and measuring the decrease in the configured predictive score. Large positive magnitudes indicate greater dependence of the fitted model on that predictor; negative values can occur and should not be interpreted as beneficial causal effects.

Task	Target	Feature set	Model	Rank	Feature	Mean importance	Relative %
regression	AC	fs1	ANN	1	PI/FF	0.446	39.877
regression	AC	fs1	ANN	2	Distance	0.420	37.541
regression	AC	fs1	ANN	3	Depth	0.178	15.903
regression	AC	fs1	DT	1	Depth	1.122	75.177
regression	AC	fs1	DT	2	Distance	0.280	18.777
regression	AC	fs1	DT	3	PI/FF	0.060	4.023
regression	AC	fs1	EN	1	PI/FF	0.558	93.448
regression	AC	fs1	EN	2	Depth	0.039	6.552
regression	AC	fs1	EN	3	Distance	0.000	0.000
regression	AC	fs1	ET	1	PI/FF	0.478	43.708
regression	AC	fs1	ET	2	Distance	0.326	29.866
regression	AC	fs1	ET	3	Depth	0.215	19.686
regression	AC	fs1	GBR	1	Depth	0.476	49.813
regression	AC	fs1	GBR	2	PI/FF	0.392	40.996
regression	AC	fs1	GBR	3	Distance	0.088	9.190
regression	AC	fs1	KNN	1	PI/FF	1.021	39.798
regression	AC	fs1	KNN	2	Distance	0.707	27.570
regression	AC	fs1	KNN	3	Depth	0.547	21.317
regression	AC	fs1	LASSO	1	PI/FF	0.484	100.000
regression	AC	fs1	LASSO	2	Distance	0.000	0.000
regression	AC	fs1	LASSO	3	Depth	0.000	0.000
regression	AC	fs1	MLR	1	PI/FF	0.918	90.010
regression	AC	fs1	MLR	2	Depth	0.067	6.532
regression	AC	fs1	MLR	3	Ucs_class	0.025	2.430

Task	Target	Feature set	Model	Rank	Feature	Mean importance	Relative %
regression	AC	fs1	RF	1	Depth	0.596	48.655
regression	AC	fs1	RF	2	PI/FF	0.380	30.994
regression	AC	fs1	RF	3	Distance	0.239	19.461
regression	AC	fs1	RIDGE	1	PI/FF	0.485	87.644
regression	AC	fs1	RIDGE	2	Depth	0.068	12.209
regression	AC	fs1	RIDGE	3	Distance	0.001	0.100
regression	AC	fs1	SVR	1	PI/FF	0.514	67.182
regression	AC	fs1	SVR	2	Distance	0.138	17.994
regression	AC	fs1	SVR	3	Depth	0.081	10.529
regression	AC	fs1	XGB	1	Depth	0.621	55.603
regression	AC	fs1	XGB	2	PI/FF	0.357	31.953
regression	AC	fs1	XGB	3	Distance	0.135	12.131

Reproducibility record

Item	Value
Python	3.13.2 (tags/v3.13.2:4f8bb39, Feb...
Platform	Windows-10-10.0.19045-SP0
Active params	G:\Researching\GT_GL_MLops\GLFS_M...
Random state	42
Regression reports	G:\Researching\GT_GL_MLops\GLFS_M...
Classification reports	G:\Researching\GT_GL_MLops\GLFS_M...
pkg:numpy	2.5.2
pkg:pandas	2.3.3
pkg:scipy	1.18.0
pkg:scikit-learn	1.9.0
pkg:xgboost	3.4.1
pkg:statsmodels	0.14.6

Item	Value
pkg:matplotlib	3.11.1
pkg:joblib	1.5.3
pkg:PyYAML	6.0.3
pkg:reportlab	5.0.1

Interpretation note

The independent test partition should be interpreted as final holdout evidence and not as a tuning criterion. Nested CV, repeated/group-aware CV, learning curves, and permutation sensitivity answer different questions and should be considered jointly rather than collapsed into a single performance number.